AI Risk Assessment
NovaHire

System: NVA-AI-003 · NovaHire

Version Assessed: v3.1 (as deployed)

Assessment Date: April 2026

Prepared by: NovaAI Labs AI Governance Office

Review Status: Draft — pending Board endorsement

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Section 1: System Overview

Nova Hire is an internally built transformer-based recruitment tool deployed by Nova AI Labs HR and licensed to 47 enterprise customers across the European Union. The system ingests CVs, job descriptions, and LinkedIn data to produce ranked shortlists with fit scores ranging from 0-100.

Deployment Timeline: October 2023 across Nova AI Labs HR operations and enterprise customers.

Provider: Nova AI Labs (internal development and maintenance).

Deployers: Nova AI Labs HR department + 47 enterprise customers.

Affected Persons: Approximately 180,000 job applicants have been processed to date.

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Section 2: Risk Assessment Methodology

This assessment uses a standard 5-point likelihood and impact scale. Risk scores are computed as the product of likelihood and impact, resulting in inherent and residual risk ratings.

Likelihood Scale (1-5):

1 = Remote (unlikely) | 2 = Low | 3 = Medium | 4 = High | 5 = Almost Certain

Impact Scale (1-5):

1 = Negligible | 2 = Minor | 3 = Moderate | 4 = Severe | 5 = Catastrophic

Risk Levels:

1-6 = Low | 7-14 = Medium | 15-19 = High | 20-25 = Critical

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Section 3: Risk Register

RISK-001 · Gender-Based Proxy Discrimination

[CRITICAL]

Description: February 2026 internal bias audit confirmed women applying for technical roles shortlisted at rates 18% lower than men with equivalent qualifications. Model appears to have learned from historical tech hiring patterns. NLP features like writing style, absence of certain keywords, career gaps correlated with gender.

Affected Groups: Women applicants, particularly for engineering and ML research roles

Likelihood: 5 | Impact: 5 | Inherent Risk: **25**

Proposed Controls: Immediate model retraining with debiased dataset; suspend automated shortlisting below-threshold rejection; independent re-audit; notify all 47 enterprise customers; engage employment law counsel

Residual Risk Score: **10** — MEDIUM

Control Owner: Chief AI Officer + Chief People Officer

RISK-002 · Age-Based Discrimination in Senior Role Shortlisting

[CRITICAL]

Description: Bias audit confirmed candidates over 45 shortlisted at 22% lower rates for senior positions. Mirrors 2024 Workday class-action. Model proxies: graduation year, years of experience framing, technology stack references.

Affected Groups: Candidates age 45+, particularly for senior/leadership roles

Likelihood: 5 | Impact: 5 | Inherent Risk: **25**

Proposed Controls: Remove graduation year and temporal markers from training features; human-mandatory review for all senior role shortlisting; legal review of age discrimination exposure; retrospective audit of past 12 months rejections

Residual Risk Score: **10** — MEDIUM

Control Owner: Chief AI Officer + General Counsel

RISK-003 · Absence of Human Oversight in Enterprise Customer Deployments

[CRITICAL]

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Description: NovaAI Labs provides no enforceable human-oversight requirements to enterprise customers. Customers may configure NovaHire to auto-reject below threshold with no human review. NovaAI Labs cannot confirm whether Article 14 oversight is being maintained in customer deployments.

Affected Groups: All 47 enterprise customers and their applicants

Likelihood: 5 | Impact: 4 | Inherent Risk: **20**

Proposed Controls: Mandatory human oversight contractual requirement; technical enforcement via product config; customer attestation; audit rights

Residual Risk Score: **8** — MEDIUM

Control Owner: Chief AI Officer + VP Enterprise Sales

RISK-004 · Inadequate Deployer Documentation (Article 13)

[CRITICAL]

Description: Enterprise customers have not been supplied with instructions for use, intended purpose, performance characteristics, known limitations, or human oversight measures as required under EU AI Act Article 13. NovaAI Labs as provider is non-compliant.

Affected Groups: All 47 enterprise deployers, regulatory authorities

Likelihood: 5 | Impact: 4 | Inherent Risk: **20**

Proposed Controls: Produce comprehensive instructions for use; publish model card; technical documentation per Annex IV; distribute to all 47 deployers

Residual Risk Score: **5** — LOW

Control Owner: Chief AI Officer + Legal

RISK-005 · Lack of Explainability for Adverse Decisions

[HIGH]

Description: Rejected candidates cannot receive meaningful explanation. Fit score only. GDPR Article 22 and EU AI Act Article 13 risk.

Affected Groups: Job applicants receiving adverse decisions

Likelihood: 5 | Impact: 3 | Inherent Risk: **15**

Proposed Controls: Implement SHAP-based explanation API; human review as decision point; update candidate communications

Residual Risk Score: **6** — LOW

Control Owner: Head of Product + Legal

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RISK-006 · Training Data Provenance Concerns

[HIGH]

Description: NovaHire was trained on a proprietary dataset including publicly scraped LinkedIn profiles. Lawful basis for processing unclear. Potential GDPR Article 6 and 14 issues. Model card incomplete.

Affected Groups: Individuals whose data was scraped without consent

Likelihood: 4 | Impact: 4 | Inherent Risk: **16**

Proposed Controls: Legal review of training data basis; produce comprehensive model card; consider retraining with consented-only dataset; engage DPO

Residual Risk Score: **8** — MEDIUM

Control Owner: Chief AI Officer + DPO

RISK-007 · Cross-Border Regulatory Exposure

[HIGH]

Description: Enterprise customers span EU, UK, and US. Multi-jurisdictional exposure: EU AI Act, UK Equality Act 2010, US state laws (NYC Local Law 144 bias audit requirements, Illinois AI Video Interview Act, Colorado AI Act 2024).

Affected Groups: Organization, regulators, customers across multiple jurisdictions

Likelihood: 4 | Impact: 4 | Inherent Risk: **16**

Proposed Controls: Multi-jurisdictional legal review; jurisdiction-specific compliance modules; regulatory monitoring programme

Residual Risk Score: **8** — MEDIUM

Control Owner: General Counsel + Compliance

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Section 4: Risk Summary Matrix

ID	Risk Title	L	I	Inherent	R	Residual
RISK-001	Gender-Based Proxy Discrimination	5	5	25	25	10
RISK-002	Age-Based Discrimination in Senior Role Shortlisting	5	5	25	25	10
RISK-003	Absence of Human Oversight in Enterprise Customer Deployments	5	4	20	20	8
RISK-004	Inadequate Deployer Documentation (Article 13)	5	4	20	20	5
RISK-005	Lack of Explainability for Adverse Decisions	5	3	15	15	6
RISK-006	Training Data Provenance Concerns	4	4	16	16	8
RISK-007	Cross-Border Regulatory Exposure	4	4	16	16	8

L = Likelihood, I = Impact, R = Residual Risk Product

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Section 5: Residual Risk Position

Overall Risk Assessment: HIGH

Despite proposed control implementations, the residual risk position remains HIGH due to the confirmed discriminatory outcomes in the February 2026 internal bias audit. Two critical risks (gender-based and age-based discrimination) have been validated through empirical evidence, indicating systemic bias in the trained model that cannot be fully mitigated through process controls alone.

Immediate Actions Required:

1. Suspend automated shortlisting decisions below rejection threshold across all deployments (internal and enterprise).
2. Notify all 47 enterprise customers of identified discriminatory bias and remediation plan.
3. Engage external employment law counsel to assess legal exposure under GDPR, EU AI Act, and domestic equality law.

90-Day Remediation Programme:

Days 1-30: Complete model retraining with debiased dataset; remove temporal and demographic proxy features; conduct independent re-audit.

Days 31-60: Distribute comprehensive instructions for use and model cards to all 47 enterprise deployers; establish human-oversight requirements in product configuration.

Days 61-90: Conduct retrospective audit of all rejections from October 2023-present; implement SHAP-based explainability API; establish AI Governance Committee for ongoing oversight.

Board Decision Required:

Board approval is required for the 90-day remediation program and for the decision to continue NovaHire deployments during the retraining period. Alternative approaches such as immediate suspension of all automated shortlisting pending model retraining should also be evaluated.

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Section 6: Recommended Human Oversight Model

Immediate (Weeks 1-4):

Mandatory human review required for 100% of shortlisting decisions, both in NovaAI Labs internal hiring and all enterprise customer deployments. System must enforce this requirement and prevent auto-rejection configurations.

Enhanced Monitoring (Weeks 5-12):

Quarterly demographic audits across all shortlists; tracking of human reviewer override rates; analysis of acceptance vs. rejection patterns by protected characteristic.

Governance & Accountability (Weeks 13+):

Establishment of AI Governance Committee with quarterly review of all bias metrics; annual independent external audit; formal escalation procedures for anomalous patterns.

This assessment is a point-in-time evaluation as of April 2026. The assessment will be reviewed and updated upon any material change to the system, training data, deployment scope, or regulatory environment.

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